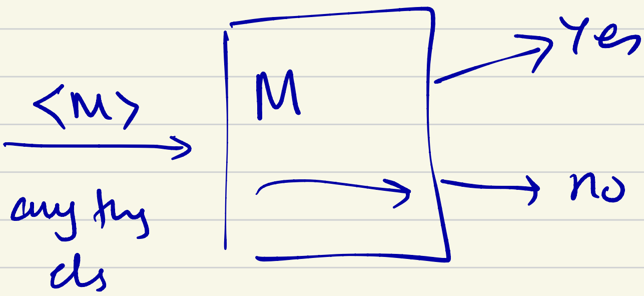
Tutorial

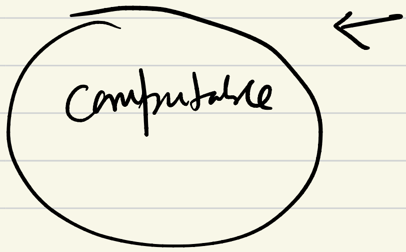
27 - April - 2024.

April 27, 2024. Tutorial Session

Let $S = \{\langle M \rangle \mid M \text{ is a TM and } L(M) = \{\langle M \rangle\}\}$.

Show that neither S nor $\bar{S}$ is Turing-recognizable.





Let B be the set of all infinite binary sequences

Show that B is uncountable using a proof by diagonalization.

$\Sigma = \{0,1\}$. $\quad \Sigma^*$ $\quad$ power set.

$\mathbb{N} = \{0, 1, 2, 3, \ldots\}$ $\quad 2^{\mathbb{N}}$

$P(\mathbb{N}) = \{ \phi, \{0\}, \{1\}, \{2\}, \ldots$

$\{0,1\}, \{0,2\}, \{0,3\}, \ldots$

$\{0,1,2\}, \{0,1,3\}, \ldots$

$\vdots$

$\}$

all possible subsets of $\mathbb{N}$.

$$\underbrace{|P(\mathbb{N})|}_{} = 2^{|\mathbb{N}|}$$

this is uncountable.

set of all infinite binary string

$B = \{$ all strings of length 0 ε

. . .   1   0, 1

. . .   2   00, 01, 10, 11

. . .   3   000, 001, . . . , 111

⋮   ⋮

0   1 2   3 4 5 6   7 8

$\}$

#	#	#	#
1 +	2 +	4 +	8
0	1	2	3

length

$$S = 2^0 + 2^1 + 2^2 + 2^3 + \dots + 2^k + \dots$$

Divergent sequence

$$\lim_{n \to k} S = 2^{k+1} - 1$$

$$\mathbb{N} = \{0, 1, 2, 3, \dots, k, k+1, \dots\}$$

Suppose B is contable.

we can list down all possible strings of B in some order. $b_1, b_2, \dots$

$B = \{b_1, b_2, \dots\}$

$b_1 = 0\ \_\ \_\ \_\ \_\ \_$

$b_2 = \_\ 0\ \_\ \_\ \_\ \_$

$b_3 = \_\ \_\ \_\ \_\ \_\ \_$

$b_4 = \_\ \_\ \_\ \_\ \_$

$\vdots\ \ = \_\ \_\ \_\ \_\ \_\ \_$

$S =$

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if A is a contable set

if B is a .. ..

then $A \cup B$ is contable set

$A \cap B$ . . .

$A \times B$, $A \times A$, $B \times A$, $B \times B$

if $\underline{n}$ is a natural number then

$\underbrace{A \times A \times \dots \times A}_{n}$ is also contable.

$P(A)$ is uncontable.

If A is contable & infinite then

$P(A)$ is uncontable.

$A = \{1, 2, 3\}$, $\quad P(A) = \{\phi, \{1\}, \{2\}, \{3\}, \{1,2\}, \{1,3\}, \{2,3\}, \{1,2,3\}\}$

$|A| = 3$, $\quad |P(A)| = 8$

If A is countable & A is infinite the $P(A)$ is uncountable.

Theorem: "A_{TM} is undecidable."

$A_{TM} = \{\langle M, w\rangle \mid M \text{ is a TM and } M \text{ accepts } w\}$.

Proof: Assume A_{TM} is decidable.

$\Rightarrow$ there exists a TM that decides A_{TM}

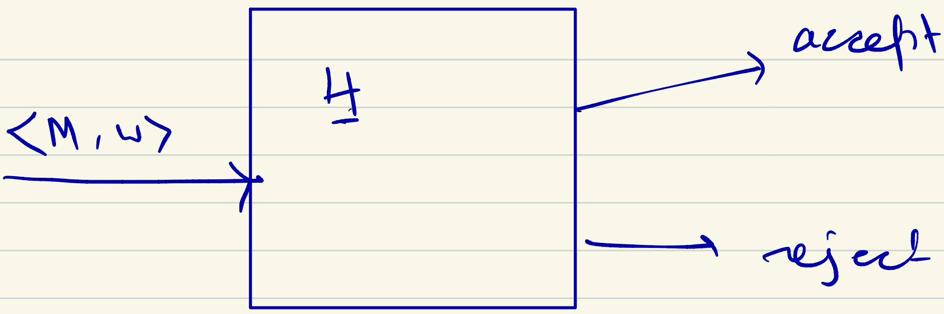
Let that TM be H. H decides A_{TM}

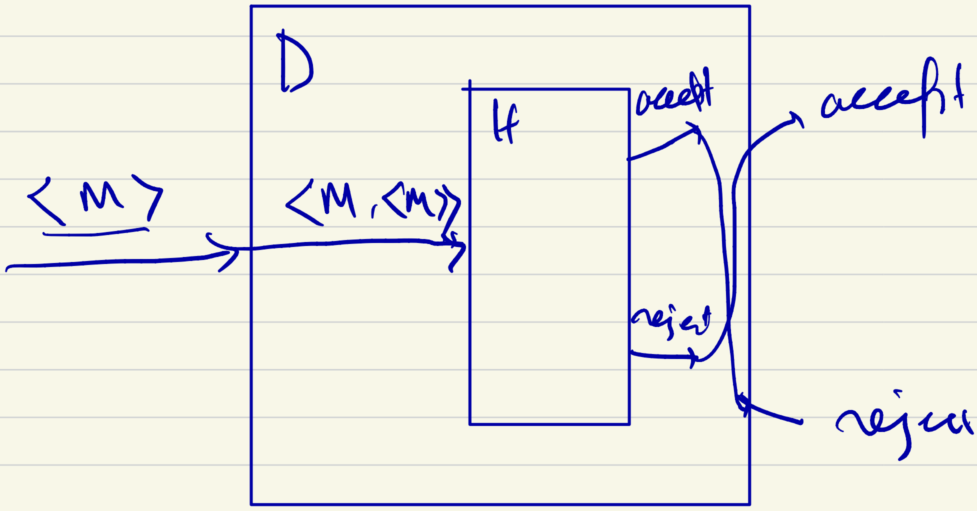
$$H(\langle M, w\rangle) = \begin{cases} \text{accept} & \text{if } M \text{ accepts } w \\ \text{reject} & \text{if } M \text{ does not accept } w. \end{cases}$$

Let us create another machine. D.

$D =$ "On input $\langle M\rangle$, where M is a TM:

1. Runs H on the input $\langle M, \langle M\rangle\rangle$
2. Output the opposite of what H returns.





$$D(\langle M\rangle) = \begin{cases} \text{accept} & \text{if } M \text{ does not accept } \langle M\rangle \\ \text{reject} & \text{if } M \text{ accepts } \langle M\rangle \end{cases}$$

$$D(\langle D\rangle) = \begin{cases} \text{accept} & \text{if } D \text{ does not accept } \langle D\rangle \\ \text{reject} & \text{if } D \text{ accepts } \langle D\rangle \end{cases}$$

contradiction.

" A is True "

Proof by contradiction: Assume A is not true then do the correct mathematical/logical steps. ⇒ Something which mathematically incorrect.

$\exists$, $\forall$

$\forall x\, P(x)$ P is true for all possible value of x in the universe of Discourse.

$\exists x\, P(x)$

$\forall x\, P(x)$ is false

$\exists x\, P(x)$ is false

Reduction

if $HALT_{TM}$ is decidable $\Longrightarrow$

A_{TM} is also decidable

E_{TM}

$REGULAR_{TM} = \{ \langle M \rangle \mid M$ is a TM and $L(M)$ is regular $\}$

undecidable
